

C-1, W-1 (Q-1)

21/06/26

"Very little management" ~ Sin

"Part of Industrial Engineering"

"वैतर्किक"

Ch - 1:

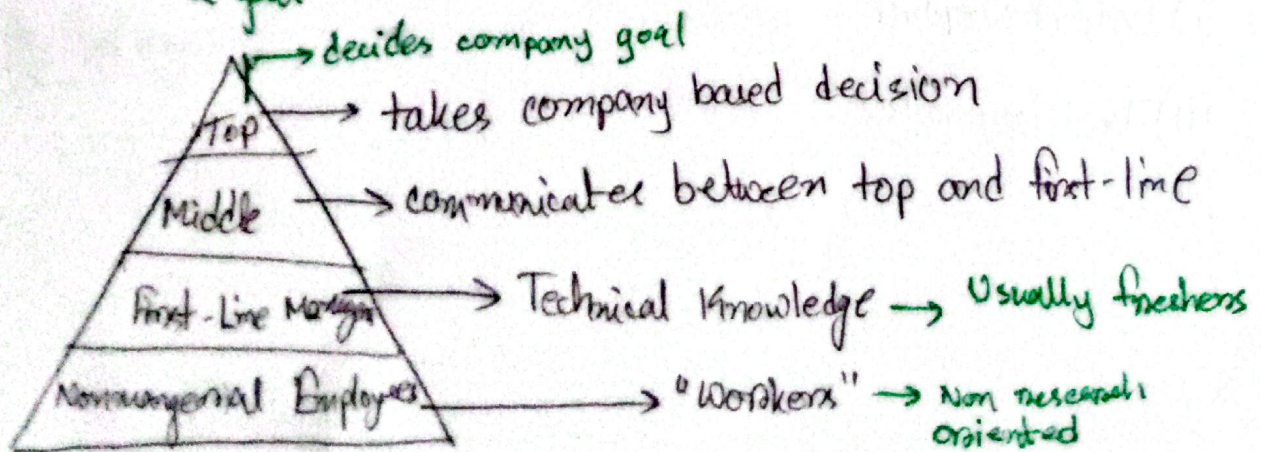
Intro. to Management

and Org → No math

\*For managerial side

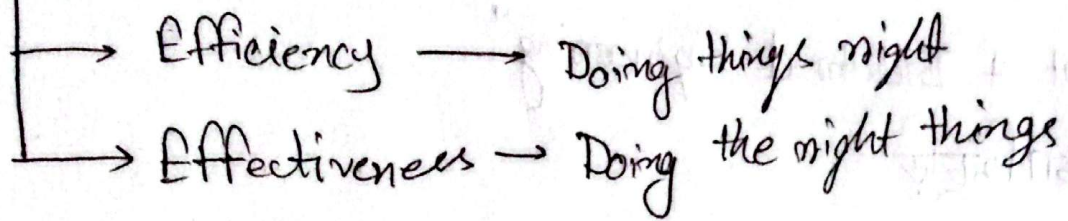
\*Managers: Someone who works with and through other people by co-ordinating their work activities in order to accomplish organizational goals

→ evaluates and mediates for accomplishment of a goal





\* Management: Efficacy



\* Functions of a Manager:

\* Planning  
\* Organizing  
\* Leading  
\* Controlling

Self-exploration  
↳ check slide if lost

\* Mintzberg's managerial roles:

- i) Interpersonal role
- ii) Informational
- iii) Decision roles



## \* Skills Approach:

i) Technical Skill

ii) Human Skill

iii) Conceptual skill / → for new situations

→ Top requires Conceptual Skill

Low level " Technical Skill

Everyone " Human Skill

Organization: Deliberate arrangement of people to accomplish some specific purpose

\* 3 Mandatory Characteristics:

- i) Have distinct purpose
- ii) Are composed of people
- iii) Have a deliberate structure

\* Better to have diff by

Challenges of a manager:

- i) Innovation
- ii) Knowledge Management

iii) Ethics

iv) Diversity

v) Globalization

vi) E-business

vii) Customers → Customer is never

satisfied  
→ so, we try to reach an acceptable level



$$c-3, w-1(a-1)$$

23/06/26



# Inventory Management

⇒ Inventory: goods or materials that a business holds for sale, production or repairs  
 ↳ raw material, <sup>work-in-</sup> ~~pro~~ <sup>cessing</sup> process & finished product + Maintenance/Repair/Operating (MRO)

\*ABC Analysis: Class A: high annual dollar volume → high rev, low sale  
Class B: medium  
Class C: low → low rev, high sale



C-4, W-2(Q-1)

28/06/26

## Holding, Ordering and Setup Cost

↓  
cost of carrying  
inventory  
(rent, labour)

↓  
cost of placing  
and receiving an order

↓  
cost of ~~some~~ preparing  
a machine for production

Dependent Demand: demand of one product increases another (razor-blade, printer-ink)

Independent Demand: ~~Opposite of dependent~~ self explanatory

→ Basic Economic Order Quantity → how much  
→ Production Order Quantity  
→ Quantity Discount Quantity



Basic EOQ: + EA  $\rightarrow$  lowest amount for purchase

\* Assumptions:

i) Demand is known, constant and independent

ii) Lead time is known and constant

$\rightarrow$  time gap between orders placement and receiving it

iii) Receipt of inventory is instantaneous and complete  
 $\rightarrow$  all info (price, lead time etc) are completed / known at the <sup>time of purchase</sup> ~~stock~~

iv) Quantity discounts are not possible  $\rightarrow$  unit price is always same

v) Only variable costs are setup / ordering and holding

vi) Stockouts can be completely avoided

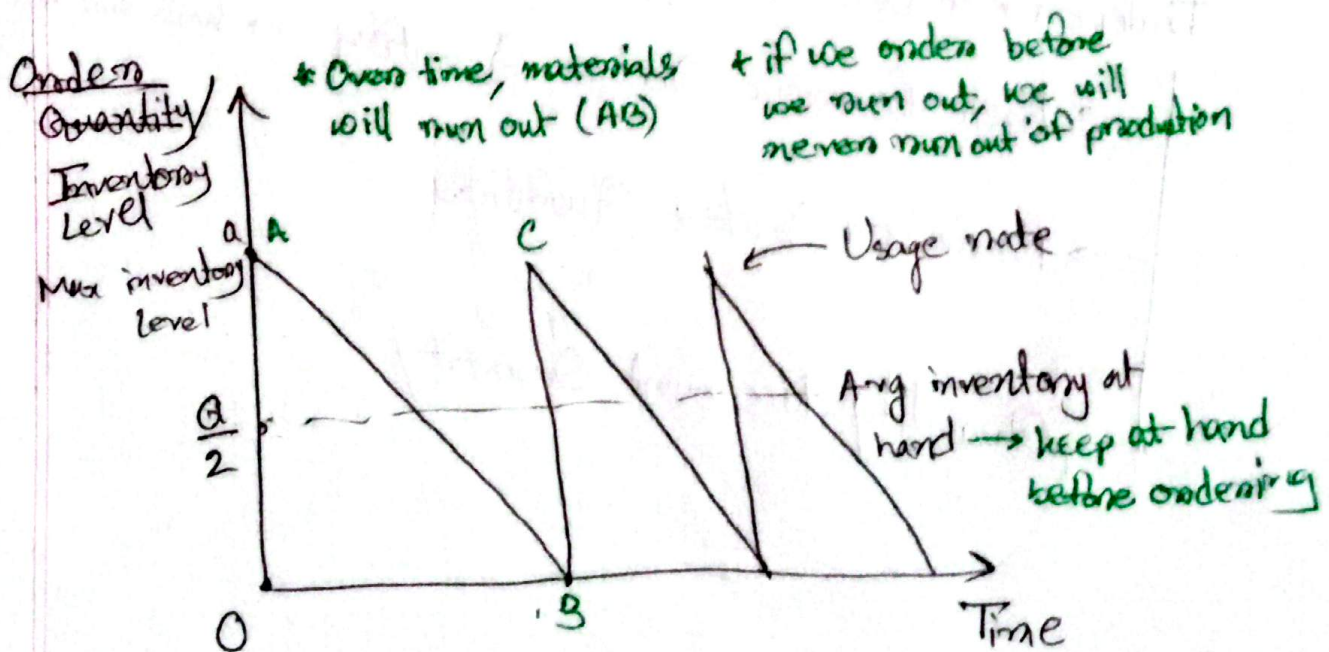
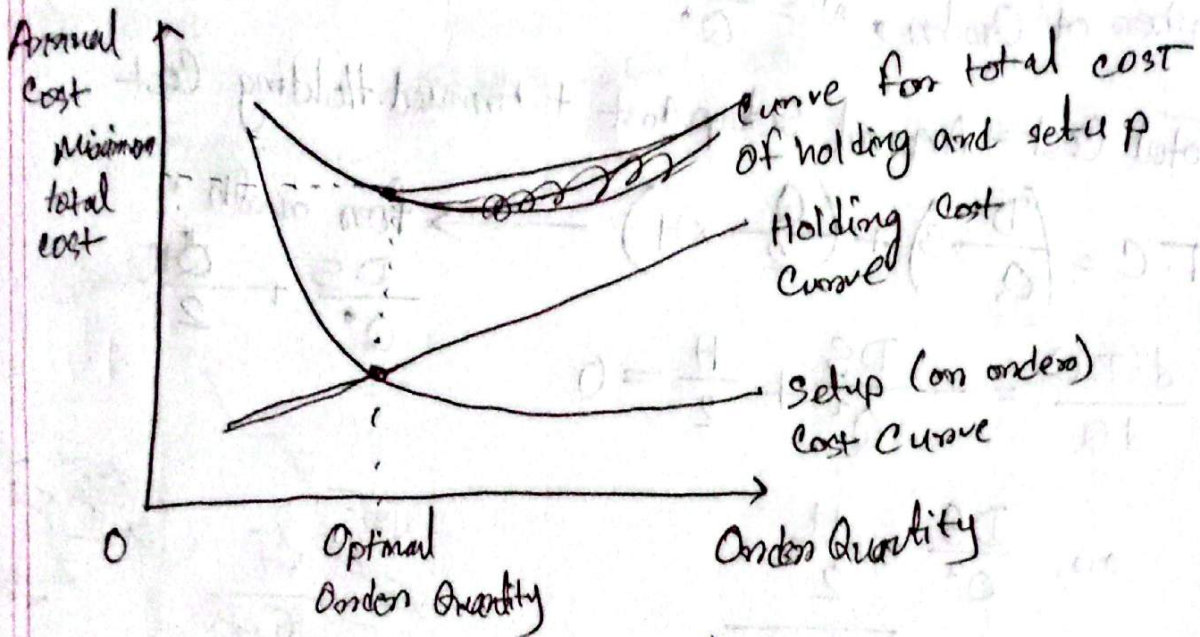


Fig: Inventory Usage Over Time



\* If we wait until material is 0, then production will remain closed  $\rightarrow$  this is bad.

\* Main Obj is to minimize cost



\*  $Q$  = Number of pieces per order

$Q^*$  = Optimal number of pieces per order

$D$  = Annual Demand in units for the Inventory item

$S$  = Setup or Ordering Cost per year

$H$  = Holding or carrying cost per unit per year

$$\begin{aligned} \text{Annual Setup Cost} &= (\text{Number of Orders placed per year}) \\ &\quad \times (\text{Setup or Order cost per Order}) \\ &= \frac{\text{Annual Demand}}{\text{Num of units in each year}} \times (\text{Setup or Or. ...}) \\ &= \left( \frac{D}{Q} \times S \right) \end{aligned}$$



Annual Holding Cost = Avg Inventory Level  $\times$  Holding Cost per unit per year

$$= \frac{Q}{2} \times H$$

Numbers of Orders,  $N = \frac{D}{Q^*}$

$\therefore$  Total Cost = Annual Setup Cost + Annual Holding Cost

$$\therefore T.C = \left( \frac{D}{Q} \times S \right) + \left( \frac{Q}{2} \times H \right) \longrightarrow \text{for math: } \frac{DS}{Q^*} + \frac{QH}{2}$$

$$\text{or, } \frac{d(T.C)}{dQ} = -\frac{DS}{Q^2} + \frac{H}{2} = 0$$

$$\text{or, } \frac{DS}{Q^2} = \frac{H}{2}$$

$$\text{or, } Q = \sqrt{\frac{2DS}{H}}$$

$$\therefore Q^* = \sqrt{\frac{2DS}{H}}$$

$$\frac{\sqrt{DS}}{DS \times H} \left( \frac{\sqrt{DS}}{H} \times H \right) = \frac{1}{\sqrt{2} \times \sqrt{DS} \times H} \times \sqrt{DS} \times H = \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{DSH}}{\sqrt{2}} + \frac{\sqrt{DSH}}{\sqrt{2}}$$

$$= \sqrt{2DSH}$$

#slide Math

+ Cycle Time : Time of ordering again

$$L \rightarrow T = \frac{\text{Total Working Day}}{N}$$



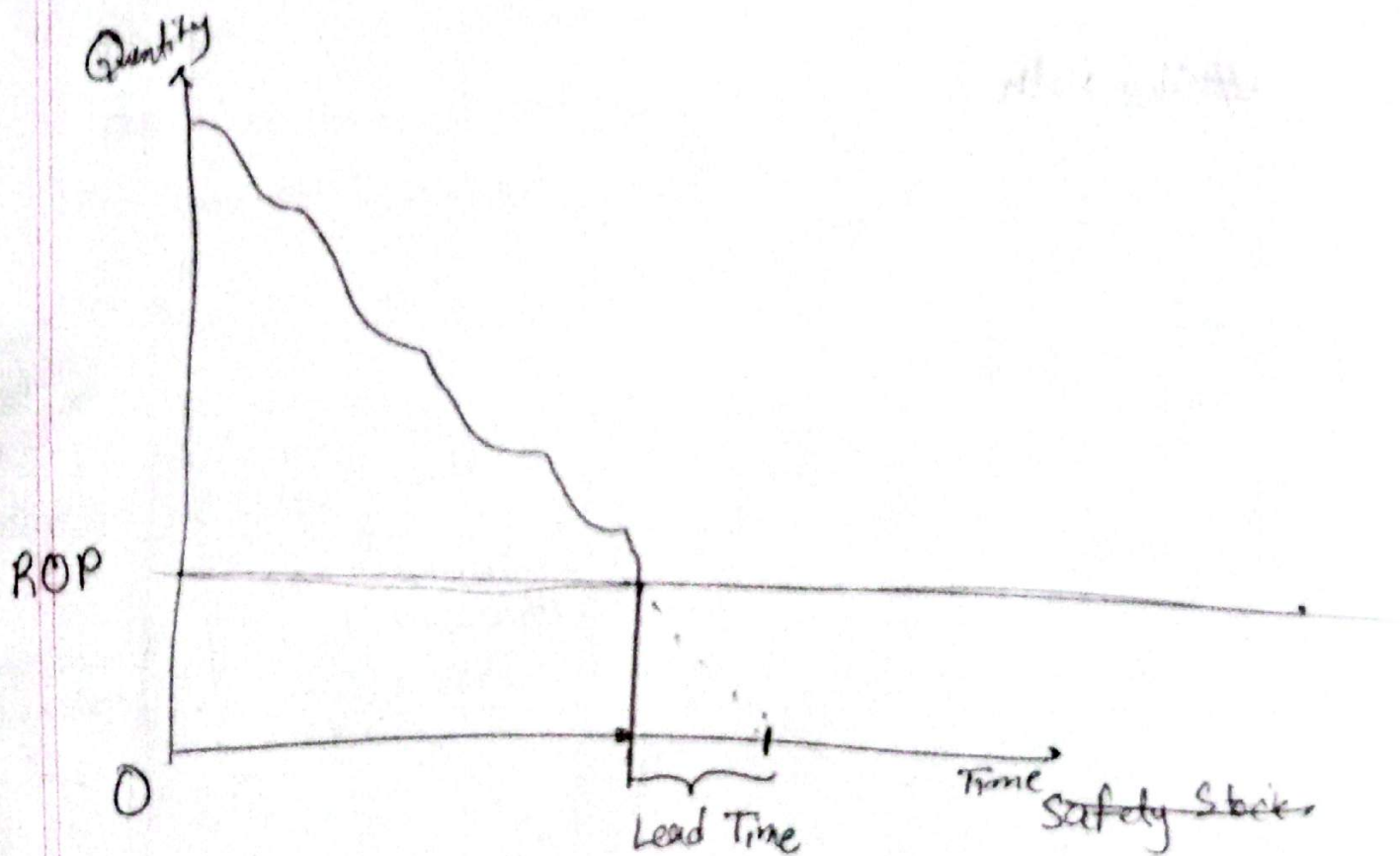
III Lead Time : . .

+ Reorder Point  $\rightarrow$  Quantity amount to reorder

+ Safety Stock  $\rightarrow$  Amount of stock kept for variable condition

+ Service Level  $\rightarrow$  Probability that demand will not exceed supply during lead time

we want more





#Reorder Point: when to order

$$\text{ROP} = (\text{Demand per Day}) \times (\text{Lead time for a new order})$$
$$= d \times L$$

$$d = \frac{D}{\text{Number of Working Days in a year}}$$

#Slide Math

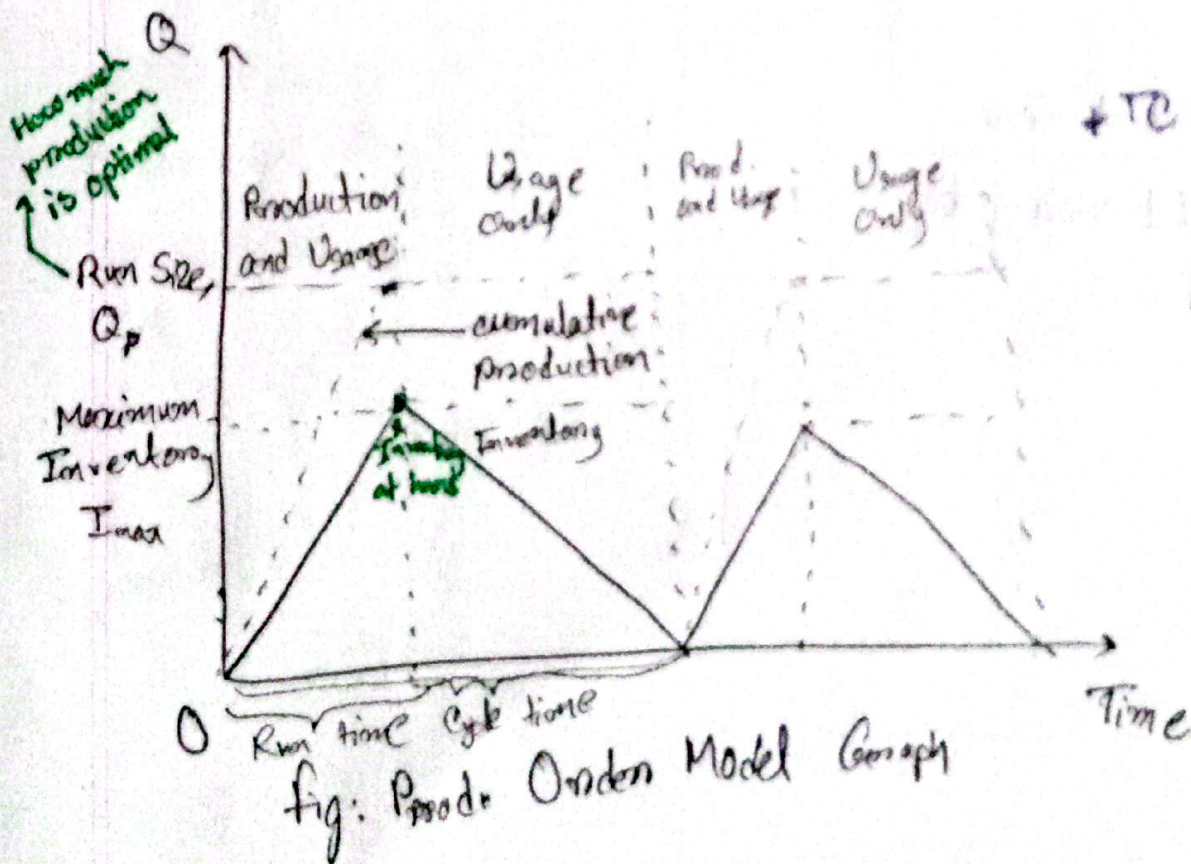


## Economic Production Quantity: perspective of a manufacturer

\* Used when units are produced and sold simultaneously

### \* Assumption

- i) Only one item is involved
- ii) Annual demand is known
- iii) Usage rate is constant  $\rightarrow$  how much is sold
- iv) Production rate is constant
- v) Lead time is does not vary
- vi) No Quantity Discounts



$$* TC = \frac{I_{max}}{2} H + \frac{DS}{Q_p}$$

$\downarrow$

$$H I_{avg} + \frac{DS}{Q_p}$$



$$Q_p = \sqrt{\frac{2DS}{H}} \times \sqrt{\frac{p}{p-u}} ; \quad p = \text{production rate} \\ u = \text{usage rate}$$

$$\text{Cycle Time} = \frac{Q_p}{u} \rightarrow \text{days after which production should start again}$$

$$\text{Run Time} = \frac{Q_p}{p} \rightarrow \text{time of production}$$

$$I_{\max} = \frac{Q_p}{p} (p-u) \quad I_{\text{avg}} = \frac{I_{\max}}{2}$$

$\downarrow$   
 Run time  $\times$   $\frac{\text{Inventory at hand}}{\text{difference of}}$

$$u = \frac{\text{use of item}}{\text{total working days}}$$



$$Q-6, n-2(Q-1)$$

(1-1) 30/06/26

# Slide Math

Quantity Discount Model: bulk = discount

$$T.C = \frac{D}{Q}S + \frac{QH}{2} + PD$$

\* Assumption:

\* Holding cost %  $\rightarrow$  % of total cost PD

# Slide Math

\* If discount price requirement  $> Q^*$ ,  
then we take the closest value

\* If discount price requirement  $< Q^*$ ,  
then we don't consider it

# Slide Math.

For deal 1,  $Q_1^* = \sqrt{\frac{2DS}{H}} = 6325 \rightarrow$  Not viable  
 $\therefore T.C_1 = \$363795$

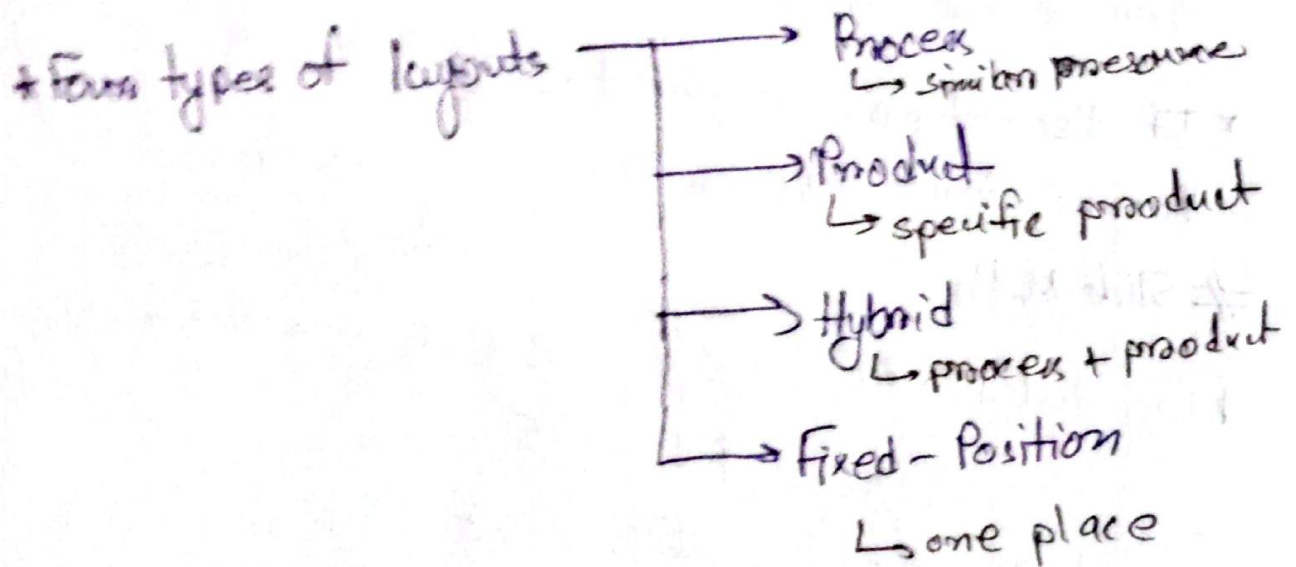
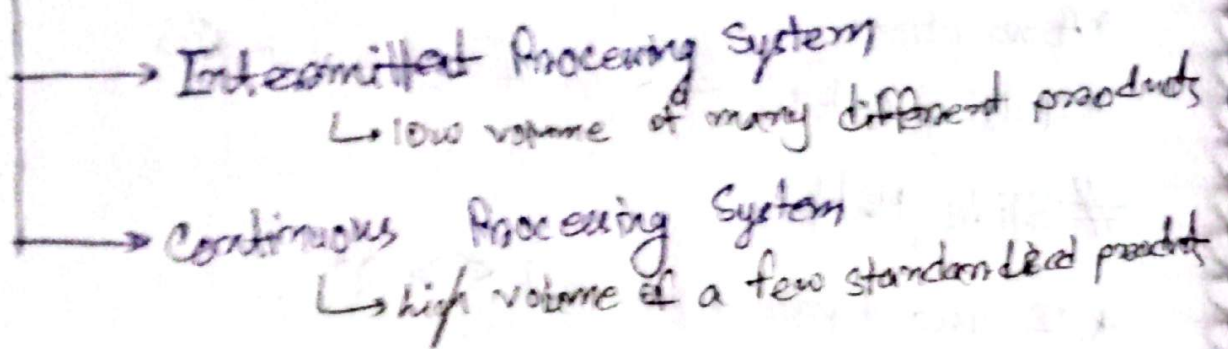
For deal 2,  $Q_2^* = \sqrt{\frac{2DS}{H}} = 6368$   
 $\therefore T.C_2 = 358970$

For deal 3,  $Q_3^* = \sqrt{\frac{2DS}{H}} = 6410 \approx 10000$   
 $\therefore T.C_3 = 354520$

$\therefore$  deal 3 is better

## Ch-10: Facility Layout → how to layout

\* Layout Planning: determining the best physical arrangement of resources





## Process Layout:

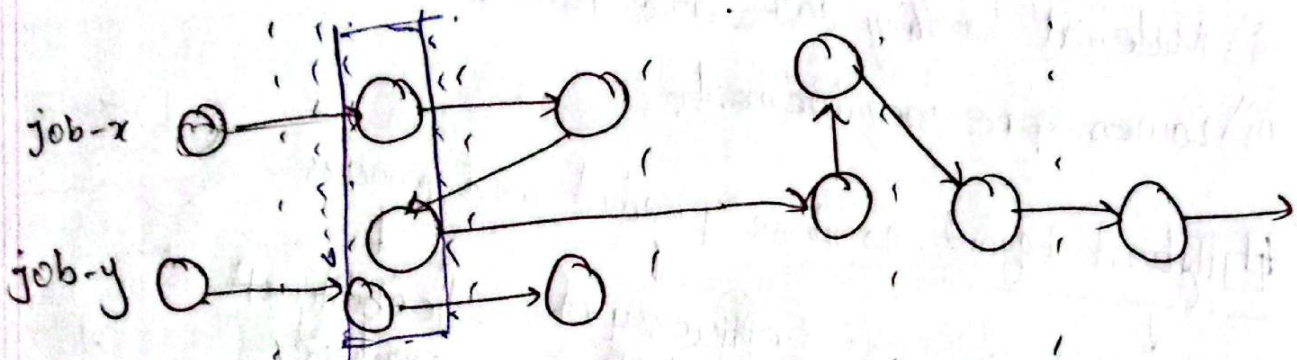
i) General purpose & flexible resources

ii)

iii)

iv) High labor intensity

\* We use ~~different~~ whatever machine is free if multiple jobs ~~each~~ require same one



## Disadvantages:

i) Processing rates are slower

ii) material handling costs are higher

iii) Scheduling resources & work flow is more complex

iv) Space requirements are higher



### Characteristics & Advantages:

### Characteristics & Advantages:

- i) Produce small numbers of products efficiently
- ii) Resources are specialized → specific machine for specific job
- iii) High capital intensity
- iv) Low flexibility relative to the market

\* Advantages:

- i) Processing rates are faster
- ii) Material handling costs are lower
- iii) Lower space requirements

Hybrid Layout: combine product and process  
↓                      ↓  
reshadowing    efficiency of production    flexibility of machines

### Fixed - position Layout:

- i) For large products
- ii) " immovable "
- iii) Resources ~~are~~ must be brought to the site
- iv) Scheduling of crews and resources is a challenge



## Cases [ ] Special Features of Layout Process Layout:

- Warehouse Layouts
- Office Layouts

[ ] Warehouse Layout: Optimizing dept locations relative to the dock

\* Crossdocking → sending products immediately instead of storing

[ ] Office Layout: Optimizing "human interaction"

[ ] Designing Product Layout → Line Balancing

→ Sequence of task, logical order, speed

\* Steps: i) Identify tasks and immediate predecessors → Make PERT

ii) Determine desired output rate

iii) Calculate cycle time  $C = \frac{\text{available time}}{\text{desired output rate}}$

iv) Theoretical minimum no. of stations →  $\text{Maximum output} = \frac{\text{available time}}{\text{cycle time}}$

v) Assign task to stations → minimize Idle time

vi) Compute efficiency  $\text{lost to the desired output}$

$$\rightarrow \frac{\sum t}{N \times C} \times 100$$

\* ch-6 for math

Quiz-1: 5th week summary,  
Layout + Inventory  
@8:15



C-9, W-3(Q-1.5)

07/07/26

#Slide math → Desired output rate = 2400 units/week, w 5 days/week, 8 hrs a day

$$r = \frac{2400 \text{ unit}}{5 \text{ days/week} \times 8 \text{ hrs/day}} = 60 \text{ unit/hr}$$

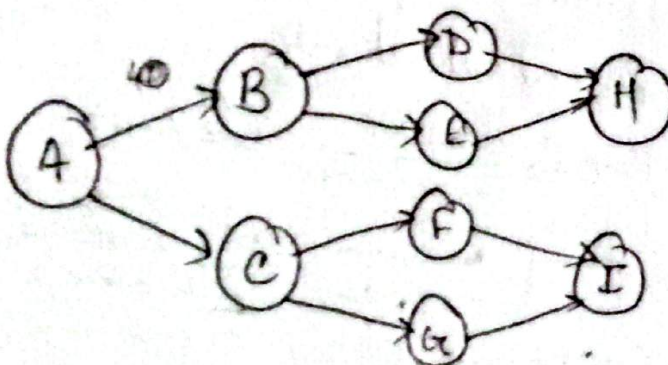
| work elem        | Time | Prec       |
|------------------|------|------------|
| A                | 40   | -          |
| B                | 30   | A          |
| C                | 50   | A          |
| D                | 40   | B          |
| E                | 6    | B          |
| F                | 25   | C          |
| G                | 15   | C          |
| H                | 20   | D, E       |
| I                | 18   | D, E, F, G |
| $\Sigma t = 244$ |      |            |

$$\therefore \text{cycle time, } c = \frac{1}{r} = \frac{1}{60} \text{ hr/unit} = 1 \text{ min/unit} = 60 \text{ sec/unit}$$

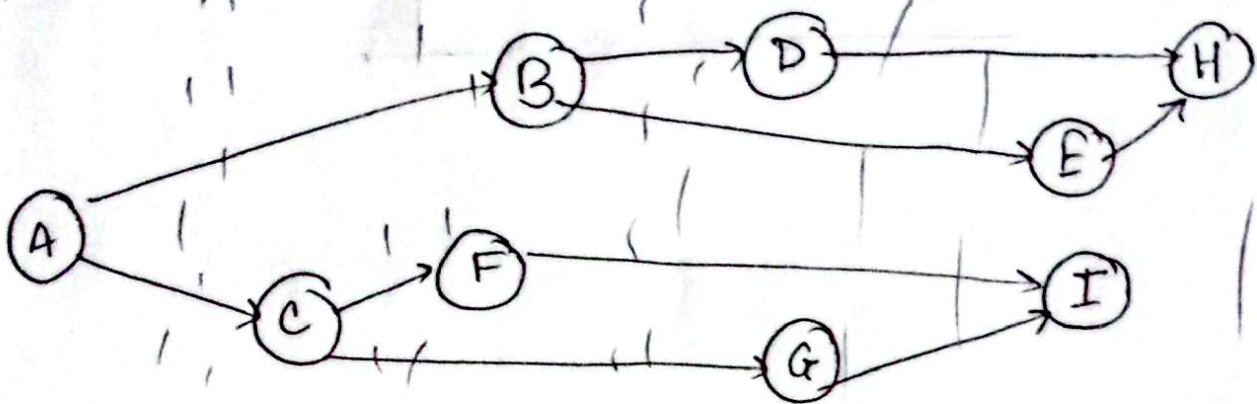
$$\therefore TM = \frac{\Sigma t}{c} = \frac{244}{60} = 4.067 = 5 \text{ stations}$$

$$\therefore \text{Idle time} = nc - \Sigma t = 5 \times 60 - 244 = 56 \text{ s}$$

$$\therefore \text{Efficiency, } e = \frac{\Sigma t}{nc} \times 100 = \frac{244}{60 \times 5} \times 100 = 81.3\%$$







1 - 2 - 1

3 - 4

5

| Currs<br>selected | Eligible <span style="color: green;">Do the slide view</span> | Selected | Avail<br><del>Id</del> |
|-------------------|---------------------------------------------------------------|----------|------------------------|
| A -               | A A                                                           | A        | 20                     |
| A                 | B, C                                                          | *        | 20                     |
| A                 |                                                               |          |                        |



## Ch - 8 : Scheduling

☐ \* Using limited resources optimally

\* 4 rules

- FCFS → First Come First Serve
- SPT → Shortest Processing Time
- EDD → Earliest Due Date
- LPT → Longest Processing Time

self explanation

Eh, close enough  
welcome  
Back OS  
Allow Time

#Slide Math:

| <u>Job Sequence</u> | Processing Time       | Cum. Time<br>↓<br>Flow Time | Job Due Date | - Due Date<br>↓<br>Job Latency |
|---------------------|-----------------------|-----------------------------|--------------|--------------------------------|
|                     |                       | ↓                           |              |                                |
|                     |                       |                             |              |                                |
|                     |                       |                             |              |                                |
|                     | Total Processing Time | Total Flow Time             |              | Total Job Latency              |

$$\text{Avg Flow Time} = \frac{\text{Total Flow Time}}{\text{No. of Jobs}}$$

$$\text{Utilization} = \frac{\text{Processing Time}}{\text{Total Flow Time}} \times 100\%$$

$$\text{Avg No. of Jobs} = \frac{\text{Total Flow Time}}{\text{Total Processing Time}}$$

$$\text{Avg latencies} = \frac{\text{Total Job Latencies}}{\text{no. of Jobs}}$$



# Johnson's Rule: N Jobs on Two Machines

# Slide Machine  
Steps:

i) Find min across <sup>both</sup> the systems

→ if (work centre 1) : send schedule first  
else: (work centre 2) " last with the next

ii) Exclude selected job and repeat (i) until all tasks are selected

→ if both are same → if diff stations: work station 1  
else: any

|             |   |   |    |    |    |    |    |
|-------------|---|---|----|----|----|----|----|
|             | 0 | 3 | 10 | 20 | 28 | 33 | 35 |
| Machine - 1 | B | E | D  | C  | A  | X  |    |
| Machine - 2 | X | B | E  | D  | C  | A  |    |
|             | 0 | 3 | 9  | 10 | 22 | 29 | 33 |

∴ Total Processing Time = 35 days

∴ Idle Time =  $(3 - 0) + (10 - 9) + (35 - 33)$   
=  $3 + 1 + 2 = 6$

DIY

|   |   |   |   |   |
|---|---|---|---|---|
| D | C | E | B | A |
|---|---|---|---|---|



N Jobs on Three Machines:

\*  $\min \{T_{1j}\} \geq \max \{T_{ij}\}$  and  $\max \{T_{1j}\} \geq \max \{T_{ij}\}$

↓                      ↓                      ↓

minimum time    maximum time  
of 1st Machine    of all other machines

→ then convert it to two machines → Machine G =  $M1 + M2$   
 " H =  $M2 + M3$

→ apply two machine job

|     | $M_1$<br>in time | $M_1$<br>out time | $M_2$<br>in | $M_2$<br>out | $M_3$<br>in | $M_3$<br>out | $M_2$<br>idle | $M_3$<br>idle |
|-----|------------------|-------------------|-------------|--------------|-------------|--------------|---------------|---------------|
| Job |                  |                   |             |              |             |              |               |               |



C-12 & 13 (w-G) Q-2

26 07126

Johnson for three:

Condition:  $\min \{ \text{machine 1} \} \geq \max \{ \text{machine 2} \}$   
or,  $\min \{ \text{machine 3} \} \geq \text{"}$  } for this method

\* We basically try to convert 3 machines to 2

\* Machine G = 1+2 } do it for two machines  
\* Machine H = 2+3 }

$L \rightarrow$  find if equal form  $H \rightarrow$  find which has  
 (1) and select it  $\rightarrow$  add at the

$(i)$   $(ii)$   $(iii)$  and  $L$   

|    |    |    |
|----|----|----|
| 13 | 10 | 15 |
| 9  | 9  | 9  |

  
 $1$   $4$   $5$ 

|   |   |   |
|---|---|---|
| 5 | 2 | 1 |
|---|---|---|

[illegible]



# Forecasting

\* Predicting future event on the basis of

- Production
- Inventory
- Personnel
- Facilities

→ use item  
time horizon  
model  
data  
validate

\* 3 types

- Short-range → 1 yr usually < 3 months  
↳ Purchasing, job scheduling, workforce levels etc
- Medium-range → 3 years > x > 3 months  
↳ Sales and production planning, budgeting
- Long-range → > 3 years  
↳ New product planning, facility location, R & D  
↳ forecasting

## Forecasting Approaches:

\* Qualitative: for vague and little data → intuition & experience  
→ opinions from experts.

\* Quantitative: for stable and historical data  
→ mathematical analysis



## Qualitative:

i) Jury of Executive Opinion: Opinions of experts

→ quick but prone to multiple thoughts but no answers

ii) Delphi Method: <sup>Anonymous</sup> survey of experts

→ Questionnaires for decision makers, staff and respondents

↳ answers discussed in meeting

→ iterative until consensus is reached.

iii) Sales Force Composite: Sales person opinions (sales reps know customers)

→ Overly optimistic

iv) Market Survey: Self explanatory

→ Sales Force but for customers

## Quantitative:

i) Time-Series: ~~Trend~~ ~~Trend~~ <sup>Short Time Existing</sup> → Pattern of → not exactly predictable

ii) Seasonal → Regular Pattern

iii) Cyclical → Repeating " → Associative & causal relationship

iv) Random → self explanatory

\* Naive Approach →  $d_1$  is true for all demand ( $d_i$ )

we don't use this

\* Moving Average → arithmetic mean →  $MA = \frac{\sum \text{demand of prev } n \text{ months}}{n}$

not widely used either

\* Weighted Moving Average → weighted mean

$$WMA = \frac{\sum (\text{weight})_n (\text{demand})_n}{\text{weight}}$$

→ still can't analyze trends

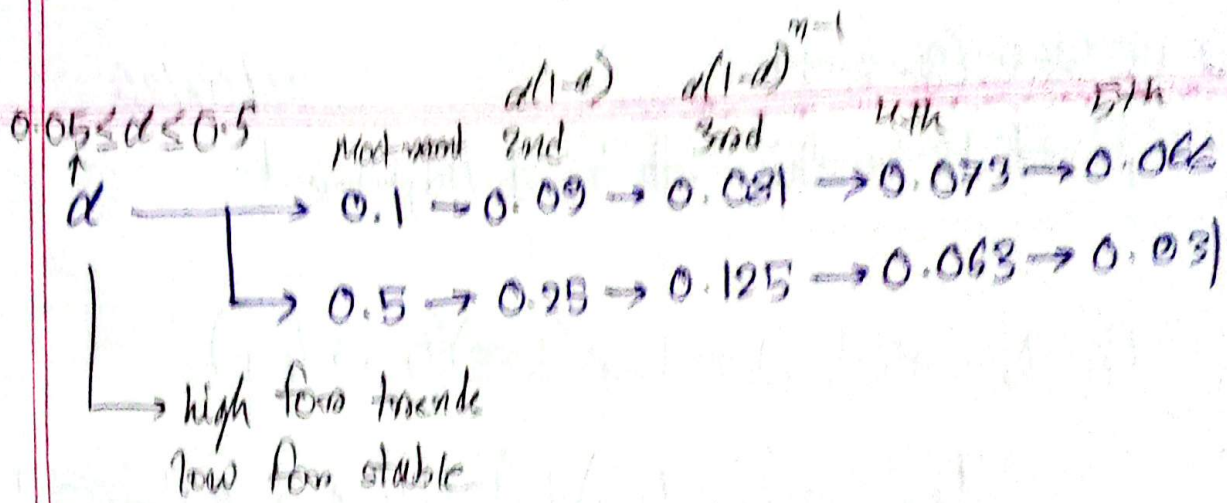
\* Exponential Smoothing → Exponentially reduces weight

→ involves little record keeping of past data

New Forecasting = Last Period's Forecast +  $\alpha$  (Actual Forecast - Last Period's Forecast)

$$F_t = F_{t-1} + \alpha (A_{t-1} - F_{t-1})$$





$$\text{Forecast Error} = F_t - A_t = F_t - F_{t-1}$$

$$\text{Mean Absolute Error} = \frac{\sum |\text{Actual} - \text{Forecasted}|}{n}$$

$$= \frac{\sum |\text{Forecast Error}|}{n}$$

$$\text{Mean Squared Error} = \frac{\sum (\text{Forecast Error})^2}{n} = \frac{(e_{m1})^2 + (e_{m2})^2 + \dots + (e_{mn})^2}{n}$$

$$\text{Mean Absolute Percent Error} = \frac{\sum 100 \times \frac{|\text{Forecast Error}|}{\text{Actual}}}{n}$$

Quiz - 2: 8th week / 7th week  
Scheduling + Forecasting

C-14, W-6, (Q-2.5)

28/07/26

\* Exponential Smoothing with Trend Adjustment:

$$F_t = \alpha(A_{t-1}) + (1-\alpha)(F_{t-1} + T_{t-1})$$

$$T_{t+1} = \beta(F_{t+1} - F_t) + (1-\beta)T_t \quad ; \quad 0 \leq \beta \leq 1$$

\* Linear Regression:  $\hat{y} = a + bx$

$$b = \frac{(\sum xy) - (n\bar{x}\bar{y})}{(\sum x^2) - (n\bar{x}^2)}$$

$$a = \bar{y} - b\bar{x}$$



C-15, W-7, (Q-2.5)

02/08/26

Managing Quality → mostly theoretical

↓  
Industry QA

Quality and Strategy: \* originated from Japan

\* Quality helps firms reduce costs

Quality  $\propto$  Price  $\propto$  Reputation ( $\propto$  Sales)

\* Quality: The totality of features and characteristics of a product or service that bears on its ability to satisfy stated or implied needs

↓  
i) User-Based:

ii) Manufacturing-Based: standards, and right time

iii) Product-Based:



ISO 9000: Standards to be maintained

↓  
International Organization of Standards

→ International recognition

[Check Slide for this chapter  
Only anecdotes will be written]

- \* Management Principle:
- i) Top management ensures quality
  - iii) Continual Improvement → Kaizen
  - iv) Involvement of people
    - ↳ Top management may not know everything → workers may know more in some aspects
  - vi) Use data and analytics to make decisions
  - viii) Think about supplier's benefit & know how supplier provides



## Cost of Quality: Cost for

i) Prevention Cost → ~~reducing~~ reducing defects

ii) Appraisal Cost → " promotion, evaluation etc

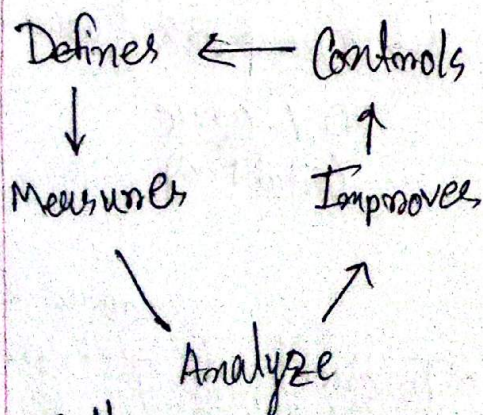
iii) External Failure Cost → " defective product/service before delivery

iv) External Failure Cost → " " " after delivery

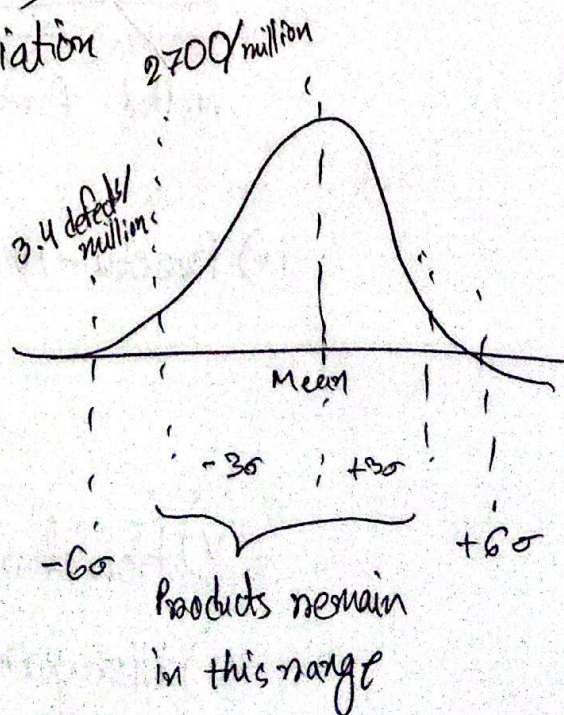
## Six Sigma: (from Motorola)

→ Sigma from Standard Deviation

### DMAIIC



→ call centre found when calls arrive more and then hired more people during that time





## Total Quality Management (TQM):

→ Use everyone to ensure long term satisfaction

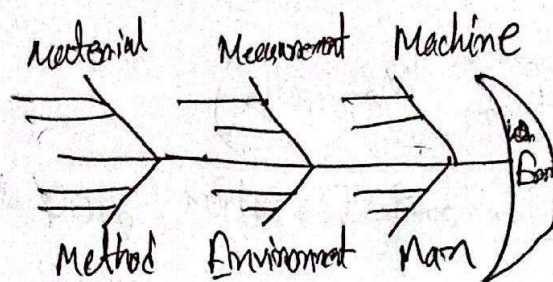
\* Basic Tools: i) Check Sheet → Hourly defect

ii) Scatter Diagram

iii) Cause - Effect Diagram / Fishbone Diagram /

Ishikawa Diagram

↳ All problems come from 5ME



↳ Method  
machine

Man

Material

Measurement  
Environment

iv) Pareto - Analysis → 80% occupies 20% of people <sup>percentage</sup>

↳ 80% problem is caused by

20% cause  
(vital few)

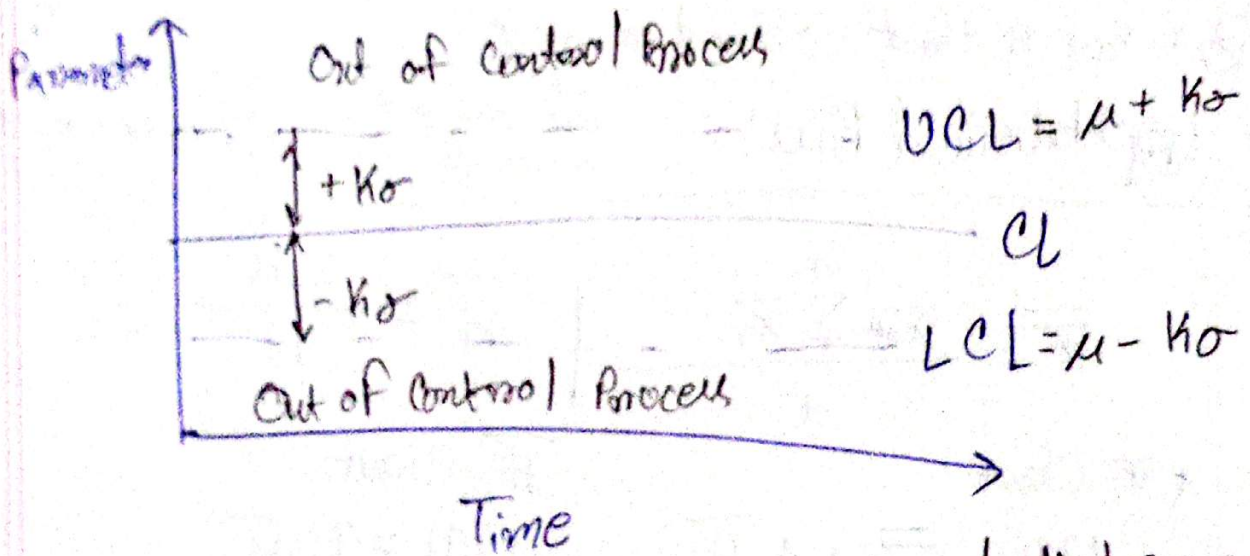
v) Flowchart

vi) Histogram → To check if <sup>normal dist</sup> histogram  
↳ if (!<sup>normal dist</sup> histogram) → find why

vii) Control Chart → Data driven analysis  
↳ find if a process is in or out of control



## Control Diagram:



- \* If equally dist bet<sup>m</sup> LCL and UCL  $\rightarrow$  controlled process
- \* If single sided (just LCL or UCL)  $\rightarrow$  out of control process  
 $\rightarrow$  we never get the desired value
- \* If trend (inc or dec) then also out of control



## Types of Control Chart:

X-bar R-Chart → Based on <sup>range</sup>

+ Statistical Bias:

↑  
Not very good  
measure

$$R = \max - \min$$

$$\bar{\bar{x}} = \frac{\sum_{i=1}^n \bar{x}_i}{n}$$

$$\bar{R} = \frac{\sum R}{N}$$

$\bar{x}$  Chart

$$UCL = \bar{\bar{x}} + A_2 \bar{R}$$

$$\text{Center line} = \bar{\bar{x}}$$

$$LCL = \bar{\bar{x}} - A_2 \bar{R}$$

R Chart

$$UCL = D_4 \bar{R}$$

$$\text{Center line} = \bar{R}$$

$$LCL = D_3 \bar{R}$$

→ Find from chart

→ if R-chart is in control, we proceed to  $\bar{x}$  chart

→ if  $\bar{x}$ -chart is in control, everything okay

\* No graph required if  
not mentioned in question

~ (^ o ^ ) ~

Thursday Week - 8 Quiz



C-18, W-8, Q-2.5  
& 19

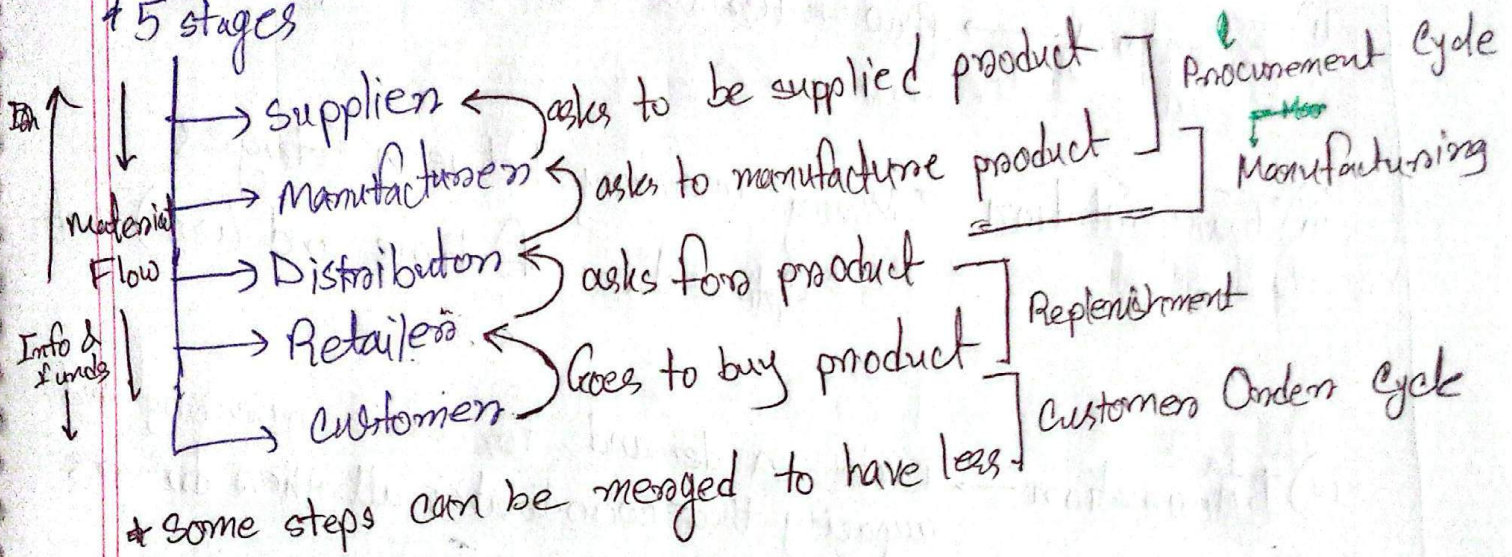
16/08/26

Mostly Business  
Related

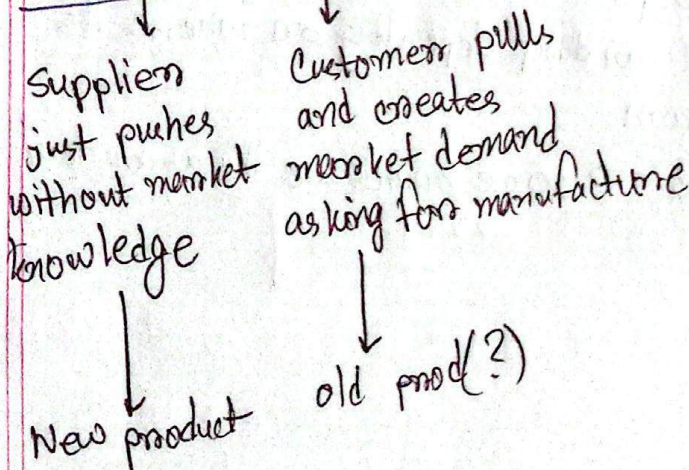
## Supply Chain Management

Supply Chain: Chain of raw material to finished product at hand

5 stages



### Push & Pull View





## Six Major Drivers of Supply Chain:

- i) Facilities → Location of Warehouse  
↳ Locations where inventory is stored, assembled or fabricated
- ii) Inventory → Raw materials and its control [prev chap]
- iii) Transportation → Moving inventory between stages  
↳ Optimization of time and (not) cost
- iv) Information → Data on demand, forecasts, inventory and capacity that coordinates all other drivers
- v) Sourcing → Deciding who performs what activity  
↳ some price can drop depending on where its ordered from
- vi) Pricing → Price of goods → some brands use this as an item



## ☐ Lack of Co-ordination:

\* Reasons → [check slide]

\* Effect → Cost ↑ Availability ↓

☐ Bullwhip Effect: Fluctuation in customer demand ( $\pm 3\%$ )  
creates high fluctuation in manufacturing ( $\pm 70\%$ )  
∴ Information flow is important

\* ERP Program

\* Causes & Countermeasures → [check slide]

Quiz shifted on Sunday



# management | Organizational Structure

## Organizational Design:

i) Work Specialization

ii) Departmentalization → Groups jobs by <sup>specialization (function, product etc)</sup> function, product etc)

iii) Chain of Command → Seniority

iv) Span of Control → No. of workers under control

v) Centralization → How decisions are taken

vi) Formalization → Rules and standards

## Common Organizational Structures: [slide]

i) Functional → Divide by background → Must be equally skilled  
under a boss → Dept Mem <sup>reports</sup> Dept Head <sup>reports</sup> . . .

ii) Matrix → Same project with multiple background

iii) Divisional → Divisions based on requirements

iv) Flat → Everyone can contact everyone → No chain of command



## Span of Control & Chain Command: [slide]

Tall

→ slow decision

Flat

→ fast decision

## Centralization & Decentralization: [slide]



# Leadership

\* for future (when management)

## Motivation Theory:

i) Maslow → Hierarchy of Need →

Self Actualization  
Esteem  
Social / Belonging  
Safety & Security  
physiological

ii) Herzberg → Two factors → Hygiene  
+ acknowledgement gives motivation → Motivators

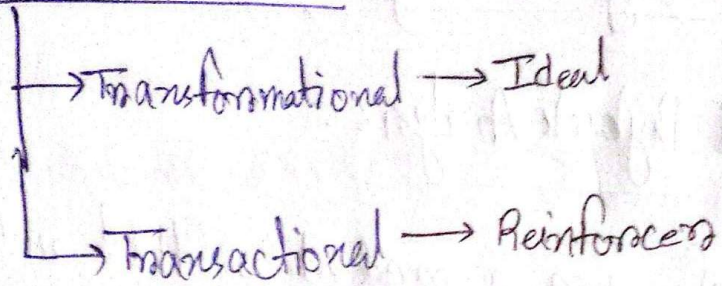
iii) McGregor → X and Y theory → X = Workers with 0 knowledge and requires control (Tactful)  
Y = Somewhat knowledgeable and can handle situations on their own (Guide)

iv) Vroom

v) McClelland



## \*Leadership Theories:





## Engineering Economy

### [H] Break-even and Payback Analysis:

[P] Break-even: Value of ~~at~~ a parameter that makes two elements equal

↓  
can be supply & demand  
revenue & cost

\* One Project bet ween two alternatives:

\* we consider linear

- Direct Solution
- Trial and Error
- Spreadsheet



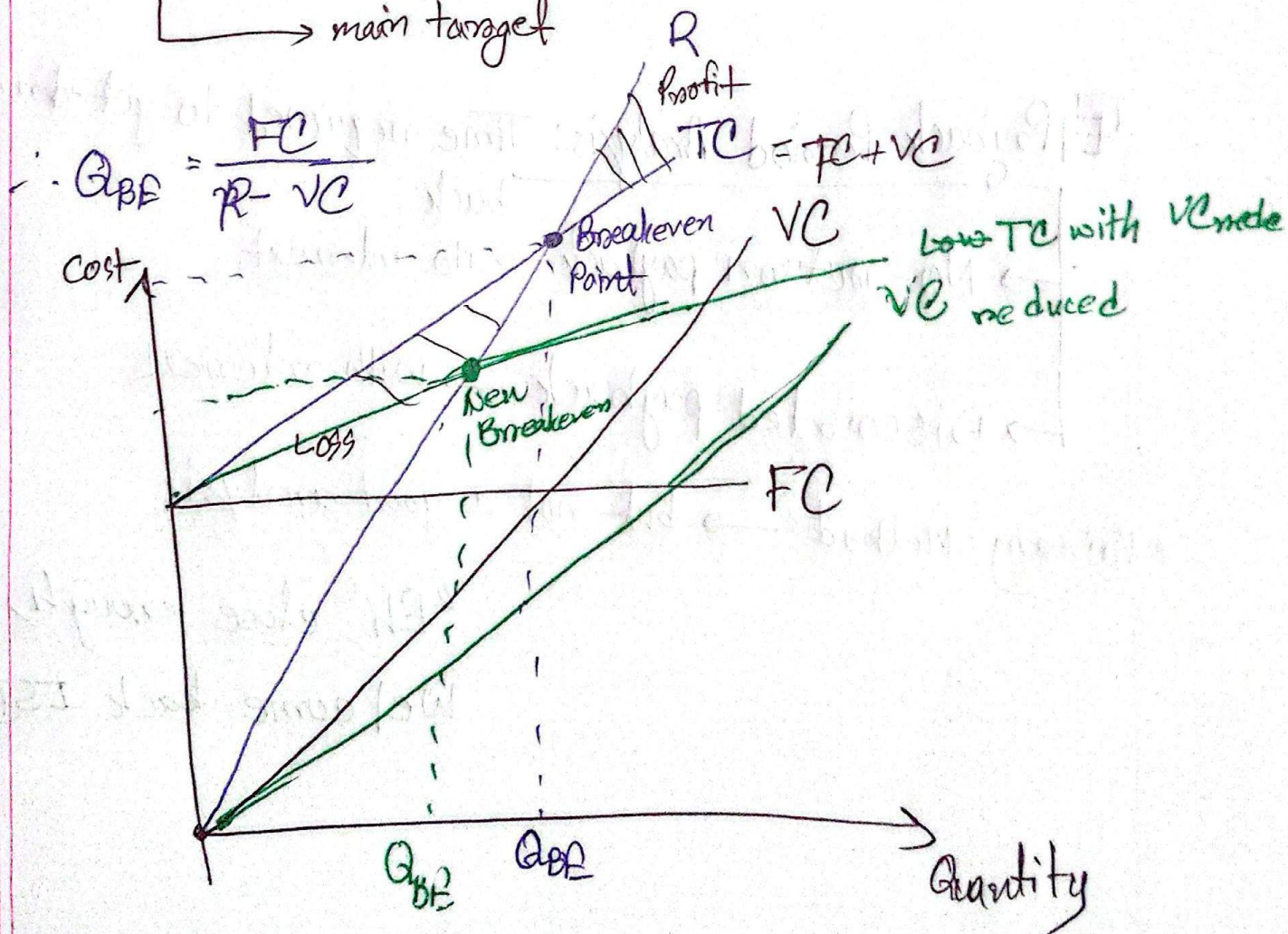
## \* Cost - Revenue Model:

- Fixed Cost: Cost ~~not directly~~ that won't change
- Variable Cost: Cost that change with parameters
- Total Cost =  $FC + VC$
- Revenue: Amount earned on quantity sold
- Profit: Amount of revenue remaining after cost

$$P = R - TC$$

$$= R - (FC + VC)$$

→ main target





### \* Between Two Alternatives:

For same  $Q_{BE}$

- if we cannot produce  $Q_{BE}$ : choose the one with lower TC  
(Alt-1 in graph)
- if we can exceed  $Q_{BE}$ : choose the one with lower TC  
(Alt-2 in graph)

Payback Period Analysis: Time required to get ~~it~~ money back

- No-return payback → No interest
- Discounted payback → with interest

\* Primary Method → but not a good analysis

"Eh, close enough,  
Welcome back ISD"



C-22, W-9, Q-3.5

25/08/26

## Introduction to Entrepreneurship

↳ MBA type shi

Quiz-3 on 11th week

~~We~~ Sunday

Presentation on 13th/14th week

Entrepreneurship: art of turning an idea into a business

\* requires multiple <sup>sources of</sup> knowledge

→ Economics is important

[Existential Crisis Time Yay!]

Please check Slide]

\* 4 main characteristics

- i) Passion for Business
- ii) Tenacity despite failure
- iii) Execution Intelligence
- iv) Product